



COURSE SLIDES

Certified Lean Six Sigma Yellow Belt (CLSSYB) Training

WSQ Course Code: TGS-2025053922

Conducted by Tertiary Infotech Academy Pte Ltd

Version 2

Welcome & Housekeeping

1

Set your mobile phone to silent mode.

2

Participate actively - no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

5

Be punctual; return from breaks on time.

6

75% attendance is required.

About the Trainer



Your Trainer

General Trainer template -
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

Digital Attendance

AM, PM and assessment attendance are mandatory.

Scan the SSG digital attendance QR code when shown.

Minimum 75% attendance is required for assessment and funding.

About the Trainer



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About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS

PhD - specialises in AI, automation and software engineering.

DELIVERS

WSQ courses on AI agents, automation (n8n) and app development.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Ground Rules

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Course Objectives

Build a Yellow Belt improvement package for Contoso Service Desk.

Practise Lean Six Sigma tools through ten aligned labs.

Prepare for assessment and certification readiness.

Learning Outcomes

Explain Yellow Belt support responsibilities.

Identify customer value, waste, defects and variation.

Use SIPOC, PDCA and DMAIC Define tools.

Plan data collection and perform basic analysis.

Support root cause, countermeasures and control planning.

Course Roadmap

Role

Value

Map

Define

Measure

Analyze

Improve

Control

Assessment

Final assessment starts Day 2 at 4:00 PM.

Assessment is completed on the LMS.

Learners should use their lab outputs and guide for revision.

Assessment Flow Reminder



The TRAQOM QR code and LMS submission are both handled through the LMS/TMS portal.

PRACTICE EXAM

Take the Online CySA+ Practice Exams

- Reinforce all four CySA+ domains with realistic, exam-style questions before assessment day.
- Use practice sets to test Security Operations, Vulnerability Management, Incident Response and Reporting.
- Review mode helps you check the correct answer and revisit the concept after each question.
- Complete the practice questions from Google Classroom, then bring difficult items to the review session.

Access and practise at:

Google Classroom > Practice Exams

The screenshot displays the 'Tertiary Exams' website interface. At the top, there are navigation links for 'Practice Exams' and 'About Us', along with 'Sign In' and a 'Get started' button. The main content area is titled 'CompTIA Cybersecurity Analyst (CySA+) Practice Exams' and includes a sub-header 'All exams / CompTIA / CySA+'. Below this, there are tabs for 'CompTIA' and 'CIS-003'. The page features three main sections: 'MODE' with a 'Practice' button, 'FOCUS' with 'CS0-003', and 'ACCESS' with a 'Class link'. A 'PRACTICE DETAILS' box on the right shows 'Exam code: C 80-003', 'Use: Review', and 'Domains: 4'. Below the main sections, there are two boxes: 'Practice mode' (Check answers and explanations while revising each domain) and 'Timed review' (Build confidence before the final assessment and Q&A). A 'FREE' box encourages starting with the classic practice set. At the bottom, a 'Domains covered' bar chart shows the following percentages: Security Operations (33%), Vulnerability Management (30%), Incident Response (20%), and Reporting & Communication (17%). A 'Learner tip' box at the bottom right suggests retaking missed items after revision.

Domain	Percentage
Security Operations	33%
Vulnerability Management	30%
Incident Response	20%
Reporting & Communication	17%

Certification and TRAQOM Survey

Complete certification and TRAQOM survey.

Confirm digital attendance records.

Keep the completed improvement package for workplace reference.

Lab 1: Yellow Belt Role, Certification Paths, and Improvement Scenario

Explain the Lean Six Sigma Yellow Belt role, compare certification path expectations, and choose a simple improvement scenario.

Contoso Service Desk wants to reduce delays in assigning support tickets. You are not leading a major Green Belt project, but you are supporting the team by mapping the process, gathering data, and testing a small PDCA improvement.

Primary skill: Yellow Belt role and readiness

Deliverable: Submit role table, certification path comparison, selected scenario, and readiness checklist.

Lab 1: Step-by-step work

Learner actions

- Step 1 - Define the Yellow Belt role
- Step 2 - Compare certification pathways
- Step 3 - Select improvement scenario
- Step 4 - Create readiness checklist

Evidence to submit

- Submit role table, certification path comparison,
- selected scenario, and readiness checklist.

Lab 1: Checkpoint

Why does a Yellow Belt usually support rather than lead major projects?

Why is process mapping an important Yellow Belt contribution?

What makes your chosen scenario suitable for a small improvement?

Lab 1: Trainer debrief

What changed in the Contoso improvement package?

Which template or table was completed?

What evidence supports the next lab?

Lab 2: Lean, Six Sigma, Waste, Voice of Customer, and Value

Identify customer value, voice of customer, waste, defects, and variation in a simple process.

Employees complain that IT support tickets are slow and status updates are unclear. The team needs to understand what customers value before choosing improvements.

Primary skill: Lean Six Sigma foundations

Deliverable: Submit VOC table, value-added table, waste walk table, and defect/variation notes.

Lab 2: Step-by-step work

Learner actions

- Step 1 - Capture voice of customer
- Step 2 - Identify value-added work
- Step 3 - Conduct a waste walk
- Step 4 - Connect Six Sigma to defects and variation

Evidence to submit

- Submit VOC table, value-added table, waste walk table, and defect/variation notes.

Lab 2: Checkpoint

What is the difference between waste and a defect?

Why should customer needs be translated into CTQs?

Which waste appears most often in your scenario?

Lab 2: Trainer debrief

What changed in the Contoso improvement package?

Which template or table was completed?

What evidence supports the next lab?

Lab 3: SIPOC, Process Mapping, Handoffs, and Yellow Belt SME Support

Create a SIPOC and process map that helps an improvement team understand boundaries, handoffs, inputs, outputs, and customers.

The service desk manager asks you to map the ticket assignment process so a Green Belt can decide whether a larger DMAIC project is needed.

Primary skill: SIPOC and process mapping

Deliverable: Submit SIPOC, detailed process map, pain point notes, and SME notes.

Lab 3: Step-by-step work

Learner actions

- Step 1 - Build SIPOC
- Step 2 - Create detailed process map
- Step 3 - Mark pain points
- Step 4 - Prepare SME notes

Evidence to submit

- Submit SIPOC, detailed process map, pain point
- notes, and SME notes.

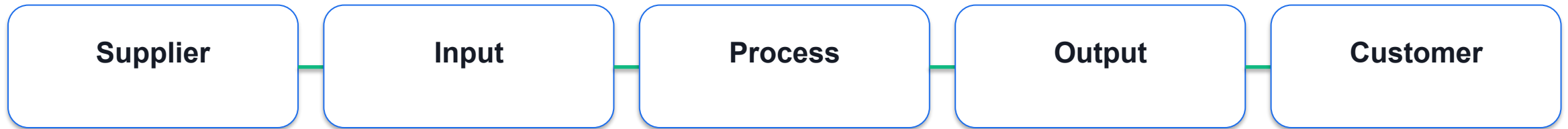
Lab 3: Checkpoint

Why is SIPOC useful before detailed mapping?

What is a handoff risk?

How can a Yellow Belt help a project leader without owning the whole project?

SIPOC turns a process into a shared map



Lab 4: PDCA Small Improvement Project Charter

Create a PDCA charter for a small improvement that a Yellow Belt could support or run under guidance.

The service desk wants to test whether a new ticket triage checklist reduces assignment delay within two weeks.

Primary skill: PDCA project charter

Deliverable: Submit PDCA table, problem statement, goal statement, scope, risk table, and test plan.

Lab 4: Step-by-step work

Learner actions

- Step 1 - Define PDCA cycle
- Step 2 - Write problem and goal statements
- Step 3 - Define scope
- Step 4 - Identify risks and assumptions

Evidence to submit

- Submit PDCA table, problem statement, goal statement, scope, risk table, and test plan.

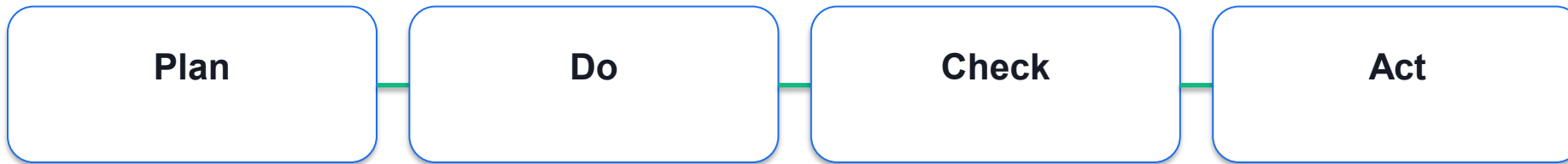
Lab 4: Checkpoint

Why is PDCA suitable for small improvements?

What makes a goal statement measurable?

What decision will you make after the Check phase?

PDCA keeps small improvements controlled



Lab 5: DMAIC Overview, Problem Statement, Scope, and Stakeholders

Support the Define phase of a DMAIC project by clarifying the problem, scope, stakeholders, customers, and expected benefits.

The PDCA pilot shows promise, but management may escalate the issue to a larger DMAIC project. Your Yellow Belt task is to prepare clear Define information.

Primary skill: DMAIC Define support

Deliverable: Submit DMAIC summary, refined problem statement, stakeholder table, project boundaries, and benefit notes.

Lab 5: Step-by-step work

Learner actions

- Step 1 - Summarize DMAIC
- Step 2 - Refine problem statement
- Step 3 - Identify stakeholders
- Step 4 - Define project boundaries

Evidence to submit

- Submit DMAIC summary, refined problem statement,
- stakeholder table, project boundaries, and benefit
- notes.

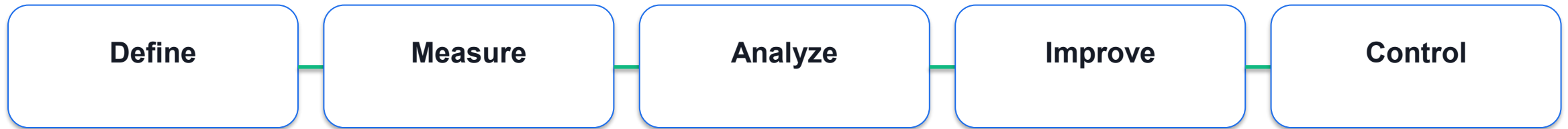
Lab 5: Checkpoint

Why should a problem statement avoid naming a solution?

Which DMAIC phases can a Yellow Belt support strongly?

Who must be involved before scope is finalized?

DMAIC gives the project a disciplined path



Lab 6: Data Collection, KPIs, Check Sheets, and Basic Metrics

Create a data collection plan with KPIs, operational definitions, check sheets, and basic metric calculations.

The service desk team has anecdotal complaints, but improvement decisions need data. You will design a simple measurement plan.

Primary skill: Measurement planning

Deliverable: Submit KPI table, check sheet, defect categories, sampling plan, and basic metric calculations.

Lab 6: Step-by-step work

Learner actions

- Step 1 - Define KPIs
- Step 2 - Create a check sheet
- Step 3 - Define defect categories
- Step 4 - Plan sampling

Evidence to submit

- Submit KPI table, check sheet, defect categories, sampling plan, and basic metric calculations.

Lab 6: Checkpoint

Why does an operational definition matter?

What bias could affect your data?

Which KPI best reflects the customer pain point?

Lab 6: Trainer debrief

What changed in the Contoso improvement package?

Which template or table was completed?

What evidence supports the next lab?

Lab 7: Pareto, Run Charts, Variation, Yield, DPU, and DPMO

Analyze simple process data using Pareto charts, run charts, variation concepts, yield, DPU, and DPMO.

You collected two weeks of service desk data. The team wants to know which defect category matters most and whether assignment time is improving.

Primary skill: Basic analysis and variation

Deliverable: Submit Pareto table, chart screenshot or notes, run chart, metric calculations, and interpretation notes.

Lab 7: Step-by-step work

Learner actions

- Step 1 - Build Pareto table
- Step 2 - Create Pareto chart
- Step 3 - Create run chart
- Step 4 - Calculate yield, DPU, and DPMO

Evidence to submit

- Submit Pareto table, chart screenshot or notes, run
- chart, metric calculations, and interpretation
- notes.

Lab 7: Checkpoint

What does a Pareto chart help prioritize?

What does a run chart show over time?

Why should you avoid overreacting to one data point?

Lab 7: Trainer debrief

What changed in the Contoso improvement package?

Which template or table was completed?

What evidence supports the next lab?

Lab 8: Root Cause Analysis with 5 Whys, Fishbone, and Evidence

Support root cause analysis using 5 Whys, fishbone diagram categories, and evidence checks.

Pareto analysis shows delayed assignment is the largest defect category. The team asks you to help identify possible causes.

Primary skill: Root cause support

Deliverable: Submit problem statement, 5 Whys table, fishbone diagram, cause priority table, and validation plan.

Lab 8: Step-by-step work

Learner actions

- Step 1 - Write the problem
- Step 2 - Complete 5 Whys
- Step 3 - Create fishbone diagram
- Step 4 - Prioritize likely causes

Evidence to submit

- Submit problem statement, 5 Whys table, fishbone diagram, cause priority table, and validation plan.

Lab 8: Checkpoint

Why should root cause analysis use evidence?

What makes a cause actionable?

Why should the team avoid jumping straight to solutions?

Root cause work moves from symptom to evidence



Lab 9: Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

Select practical countermeasures using 5S, mistake proofing, visual management, standard work, and Kaizen planning.

Root cause analysis suggests delayed assignments happen because ticket categories are inconsistent and required information is missing. The team wants low-risk countermeasures.

Primary skill: Improve tools and kaizen

Deliverable: Submit countermeasure table, priority scoring, 5S table, mistake-proofing design, and standard work checklist.

Lab 9: Step-by-step work

Learner actions

- Step 1 - Generate countermeasures
- Step 2 - Prioritize countermeasures
- Step 3 - Apply 5S thinking
- Step 4 - Design mistake proofing

Evidence to submit

- Submit countermeasure table, priority scoring, 5S
- table, mistake-proofing design, and standard work
- checklist.

Lab 9: Checkpoint

What is the difference between a countermeasure and a permanent solution?

How can 5S apply to digital work?

Why is standard work important after improvement?

Lab 9: Trainer debrief

What changed in the Contoso improvement package?

Which template or table was completed?

What evidence supports the next lab?

Lab 10: Control Plan, A3 Summary, Handover, and Certification Readiness

Create a control plan, A3-style improvement summary, handover checklist, and personal certification readiness plan.

The PDCA test improved assignment time. The team needs a simple control plan so the gain is sustained and management can decide whether to expand the solution.

Primary skill: Control and certification readiness

Deliverable: Submit control plan, visual management notes, A3 summary, handover checklist, and certification readiness plan.

Lab 10: Step-by-step work

Learner actions

- Step 1 - Create control plan
- Step 2 - Define visual management
- Step 3 - Write A3 summary
- Step 4 - Prepare handover

Evidence to submit

- Submit control plan, visual management notes, A3
- summary, handover checklist, and certification
- readiness plan.

Lab 10: Checkpoint

Why does an improvement need a control plan?

What should happen if a metric falls outside target?

Which Yellow Belt topic needs your most review?

Lab 10: Trainer debrief

What changed in the Contoso improvement package?

Which template or table was completed?

What evidence supports the next lab?

Integrated Package Review

Role table and readiness checklist

VOC, CTQ, waste and process map

PDCA/DMAIC charter and stakeholder view

Measurement and analysis evidence

Root cause, countermeasures and control plan

Final Readiness Checklist

Can you explain the Yellow Belt role?

Can you trace VOC to CTQ and metric?

Can you explain your selected root cause with evidence?

Can you describe how control will be sustained?

THANK YOU

Closing

Submit final lab deliverables.

Complete assessment and survey steps.

Apply one small improvement at work.